TASK-5

[1]
```
import pandas as pd
import numpy as np
```

[2]
```
df = pd.read_csv("/content/sample_sales_data.csv")
```

[3]
```
df
```

	Date	City	Category	Daily_Sales	Units_Sold
0	2026-03-01	New York	Electronics	1200	12
1	2026-03-02	Los Angeles	Clothing	850	17
2	2026-03-03	Chicago	Electronics	1500	10
3	2026-03-04	New York	Home	450	9
4	2026-03-05	Chicago	Clothing	950	19
5	2026-03-06	Los Angeles	Electronics	2100	15
6	2026-03-07	New York	Clothing	600	12
7	2026-03-08	Chicago	Home	1100	11
8	2026-03-09	Los Angeles	Electronics	1750	14
9	2026-03-10	New York	Home	800	16

[4]
```
sales_array = df['Daily_Sales'].to_numpy()

print("Daily_Sales_Array: ",sales_array)
```

```
Daily_Sales_Array:  [1200  850 1500  450  950 2100  600 1100 1750  800]
```

[5]
```
total_sales = df['Daily_Sales'].sum()
min_sales = df['Daily_Sales'].min()
max_sales = df['Daily_Sales'].max()
avg_sales = df['Daily_Sales'].mean()

print("Total_Sales: ",total_sales)
print("Min_Sales: ",min_sales)
print("Max_Sales: ",max_sales)
print("Average_Sales: ",avg_sales)
```

```
Total_Sales:  11300
Min_Sales:  450
Max_Sales:  2100
Average_Sales:  1130.0
```

[6]
```
df.head()
```

	Date	City	Category	Daily_Sales	Units_Sold
0	2026-03-01	New York	Electronics	1200	12
1	2026-03-02	Los Angeles	Clothing	850	17
2	2026-03-03	Chicago	Electronics	1500	10
3	2026-03-04	New York	Home	450	9
4	2026-03-05	Chicago	Clothing	950	19

Next steps: Generate code with df | New interactive sheet

[7]
```
df.tail()
```

	Date	City	Category	Daily_Sales	Units_Sold
5	2026-03-06	Los Angeles	Electronics	2100	15
6	2026-03-07	New York	Clothing	600	12
7	2026-03-08	Chicago	Home	1100	11
8	2026-03-09	Los Angeles	Electronics	1750	14
9	2026-03-10	New York	Home	800	16

[8]
```
df.shape
```
```
(10, 5)
```

[9]
```
df.columns
```
```
Index(['Date', 'City', 'Category', 'Daily_Sales', 'Units_Sold'], dtype='object')
```

[10]
```
df.info()
```
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 5 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Date         10 non-null     object
 1   City         10 non-null     object
 2   Category     10 non-null     object
 3   Daily_Sales  10 non-null     int64
 4   Units_Sold   10 non-null     int64
dtypes: int64(2), object(3)
memory usage: 532.0+ bytes
```

[11]
```
df.describe()
```

	Daily_Sales	Units_Sold
count	10.000000	10.00000
mean	1130.000000	13.50000
std	519.722148	3.24037
min	450.000000	9.00000
25%	812.500000	11.25000
50%	1025.000000	13.00000
75%	1425.000000	15.75000
max	2100.000000	19.00000

```
df.isnull().sum()
```

	0
Date	0
City	0
Category	0
Daily_Sales	0
Units_Sold	0

dtype: int64

```
df.duplicated().sum()
```
```
np.int64(0)
```

```
df['Revenue_per_unit'] = df['Daily_Sales'] / df['Units_Sold']

df
```

	Date	City	Category	Daily_Sales	Units_Sold	Revenue_per_unit
0	2026-03-01	New York	Electronics	1200	12	100.0
1	2026-03-02	Los Angeles	Clothing	850	17	50.0
2	2026-03-03	Chicago	Electronics	1500	10	150.0
3	2026-03-04	New York	Home	450	9	50.0
4	2026-03-05	Chicago	Clothing	950	19	50.0
5	2026-03-06	Los Angeles	Electronics	2100	15	140.0
6	2026-03-07	New York	Clothing	600	12	50.0
7	2026-03-08	Chicago	Home	1100	11	100.0
8	2026-03-09	Los Angeles	Electronics	1750	14	125.0
9	2026-03-10	New York	Home	800	16	50.0

[15] ✓ 0s

```
min_revenue = df['Revenue_per_unit'].min()
max_revenue = df['Revenue_per_unit'].max()
avg_revenue = df['Revenue_per_unit'].mean()
std_revenue = df['Revenue_per_unit'].std()

print("Min Revenue Per Unit: ",min_revenue)
print("Max Revenue Per Unit: ",max_revenue)
print("Avg Revenue Per Unit: ",avg_revenue)
print("Std Revenue Per Unit: ",std_revenue)
```

```
Min Revenue Per Unit:  50.0
Max Revenue Per Unit:  150.0
Avg Revenue Per Unit:  86.5
Std Revenue Per Unit:  41.368936279397964
```

[16] ✓ 0s

```
high_sales = df[df['Daily_Sales']>1000]
print(high_sales)
```

```
         Date         City     Category  Daily_Sales  Units_Sold  \
0  2026-03-01     New York  Electronics         1200          12
2  2026-03-03      Chicago  Electronics         1500          10
5  2026-03-06  Los Angeles  Electronics         2100          15
7  2026-03-08      Chicago         Home         1100          11
8  2026-03-09  Los Angeles  Electronics         1750          14

   Revenue_per_unit
0             100.0
2             150.0
5             140.0
7             100.0
8             125.0
```

[18] ✓ 0s

```
revenue_by_city = df.groupby("City")['Revenue_per_unit'].sum()
print(revenue_by_city)
```

```
City
Chicago        300.0
Los Angeles    315.0
New York       250.0
Name: Revenue_per_unit, dtype: float64
```

[19] ✓ 0s

```
ny_electronics = df[(df['City']=='New York') & (df['Category']=='Electronics')]
print(ny_electronics)
```

```
         Date      City     Category  Daily_Sales  Units_Sold  \
0  2026-03-01  New York  Electronics         1200          12

   Revenue_per_unit
0             100.0
```